HRV-Based Asthma Monitoring Project

Internship Work Report, Contribution, and Four-Month Plan

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Continued project involvement: June 2026–Present
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1. Overall Project Progress

Project goal

The project studies whether longitudinal wearable signals can characterise patient state and support early warning of asthma deterioration. The main signal studied so far has been Heart Rate Variability (HRV), alongside heart rate, temperature, SpO2, steps, and sleep. The long-term goal is to move from signal understanding to change-point detection and then to predictive modelling of asthma exacerbations.

Data and cohort analysis completed

The dataset contains approximately 12.2 million heart-rate measurements and approximately 1.0 million HRV measurements. The available signals were studied through cohort-level and patient-level summaries, including ranges, distributions, observation counts, means, medians, standard deviations, missingness, and sampling frequency.

The analysis confirmed that wearable data does not arrive on a perfectly regular grid. HRV is nominally sampled around every 10 minutes, but actual observations are irregular and affected by gaps. Missingness was studied patient by patient and across the cohort, and a 180-minute threshold was adopted, after analysis and domain input, to separate ordinary irregularity from long missing periods. This is important because later sequential methods should not treat long gaps as continuous physiology.

Main HRV finding

The key question was whether HRV moves around one fixed patient baseline or whether the normal level changes through the day. I checked this manually across all available patients using code, using tumbling-window summaries such as tumbling mean and tumbling median to inspect local HRV over time.

The same broad within-day pattern was visible for every patient checked. It was not only a pooling effect and was not only caused by one estimator. The modelling implication is simple: a single fixed HRV baseline is not enough. Future CPD and prediction should use a time-aware baseline, and later should also account for physiological context such as sleep, activity, or patient state.

Smoothing, denoising, and CPD progress

Smoothing and denoising methods were investigated before deciding whether they should be part of the pipeline. The methods considered included moving average, exponential smoothing, Kalman/state-space methods, Gaussian processes, particle filters, regime-switching methods, robust filters, spectral methods, VAEs, and related neural approaches. The main conclusion was that aggressive smoothing can remove useful physiological structure along with noise, so smoothing was not kept as a mandatory preprocessing step.

Change-point detection work was narrowed to four useful method families: Bayesian online CPD, kernel CPD, Kalman/state-space methods, and HMMs. CPD implementation was started, then paused because the full-cohort and multi-factor analysis needed to be completed first.

2. My Specific Contribution and Basis for Revised Stipend

Specific technical contributions

My role has combined implementation, statistical checking, patient-level analysis, and modelling direction. The main contributions are:

- Implemented and evaluated candidate smoothing methods for HRV.
- Investigated whether smoothing methods preserved useful temporal structure instead of only reducing variance.
- Investigated VAE-based denoising and implemented a convolutional VAE on HRV data.
- Contributed to implementations involving BOCPD, kernel CPD, Kalman/state-space methods, and HMMs.
- Checked the statistical and method assumptions behind the techniques being used.
- Verified that statistical code outputs matched the quantities they were supposed to compute.
- Carried out patient-level and full-cohort HRV analysis.
- Manually checked the within-day HRV feature across all available patients using code.
- Showed that the recurring HRV structure appears under both tumbling mean and tumbling median.
- Connected this result to the next modelling decision: later methods should not depend on a single fixed baseline.
- Continued exploratory work on predictive modelling architectures for future exacerbation prediction.

Why this contribution is valuable for the next phase

The project has now moved beyond isolated internship tasks. The next phase depends on continuity because the person doing the work needs to understand the data, the code, the patient-level patterns, the missingness, and the modelling assumptions.

- My work has already connected implementation, statistical checking, cohort analysis, and modelling direction.
- The HRV baseline finding directly affects how CPD and prediction should be designed.
- Continuing in the role would preserve technical continuity while the project moves from analysis to deliverables.

Why approximately 35 hours per week are needed

The next stage cannot be handled through occasional work. It covers multiple physiological factors, each of which needs cleaning, validation, cohort summaries, patient-level checks, interpretation, and documentation. CPD and prediction also require repeated cycles of running methods, checking outputs, correcting issues, rerunning, and comparing results with exact or approximate exacerbation dates.

- A 35-hour weekly commitment is needed for sustained progress across analysis, implementation, validation, and writing.
- I plan to distribute this time flexibly across weekdays and weekends.
- From 23 August onward, I will also be available to work in person on campus.

Basis for revised contract and stipend

My original internship contract covered 15 April 2026 to 14 June 2026. Since then, I have continued working on the HRV project under the guidance of Dr Mayank and Jatin Dangi, and I am now able to take a more active role in the project.

The proposed extension involves increased hours, broader responsibility, and direct ownership of important next-stage deliverables. A revised contract and stipend adjustment would reflect the actual time commitment, continuity, and technical contribution required for the next phase.

3. Plan for the Next Four Months

Month 1: Full-Cohort Validation and Preparation

I expect to work with Isha Kala and Jatin Dangi, under the guidance of Dr Mayank, Dr Naveen, and Dr Pratyush.

- Complete cohort-level analysis for every available physiological factor.
- Identify ghost data, implausible readings, duplicated observations, and other artefacts.
- Define factor-specific rules for detecting and removing unreliable measurements.
- Validate the cleaned data across the cohort.
- Prepare summary statistics, figures, and diagrams required for the planned cohort paper.
- Target deadline based on current progress: 15 September 2026.
- Deliverable: cleaned and statistically characterised full-cohort data with material ready for manuscript preparation.

Month 2: Factor-Specific Change-Point Detection

I expect to work with Jatin Dangi, under the guidance of Dr Mayank and Dr Pratyush.

- Define what counts as a meaningful change for each physiological factor.
- Select suitable CPD methods from Jatin Dangi's existing method collection.
- Use complementary methods where needed, including BOCPD, kernel CPD, Kalman/state-space methods, and HMMs.
- Apply selected CPD methods across the full cohort.
- Align detected changes with known exact or approximate asthma exacerbation dates.
- Continue exploring candidate machine-learning architectures for later prediction.
- Deliverable: cohort-wide candidate change annotations linked, where possible, to known exacerbation events.

Month 3: CPD Validation and Preprint

I expect to continue working with Jatin Dangi, under the guidance of Dr Mayank and Dr Pratyush.

- Validate the selected CPD framework at cohort level.
- Measure the proportion of known exacerbations associated with detected change points.
- Measure false negatives, false positives, detection timing, and consistency across patients and factors.
- Check whether the framework can achieve very high sensitivity to known exacerbations.
- Make any strong claim only if supported by the final statistical evaluation.
- Prepare the CPD methodology, selection criteria, and cohort-level results as a preprint.
- Deliverable: validated CPD results and a draft/preprint-ready report.

Month 4: Predictive Model Evaluation

- Implement and compare candidate predictive modelling architectures.
- Define training, validation, and test procedures.
- Tune model choices, hyperparameters, prediction targets, and prediction horizons.
- Evaluate performance across patients.
- Analyse false positives, false negatives, calibration, and time-to-event behaviour.
- Improve architectures and evaluation strategy based on observed results.
- Deliverable: reproducible predictive framework for estimating future exacerbation risk from patient history.